

Google Cloud Certified

Generative AI Leader

Classroom Certification Study Notes

55 Questions	90 min Time Limit	\$99 Exam Fee	70% Pass Mark
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Domain 1 Gen AI Fundamentals ~30%	Domain 2 Google Cloud AI ~35%	Domain 3 Optimization Techniques ~20%	Domain 4 Business Strategy ~15%
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No prerequisites required | Valid 3 years | Online proctored or test centre | Retake after 14 days

DOMAIN 1 | Fundamentals of Generative AI (~30% of exam)

1.1 Core Gen AI Definitions

Term	Official Definition
Artificial Intelligence (AI)	Building machines that can perform tasks that typically require human intelligence, such as learning, problem-solving, and decision-making
Machine Learning (ML)	A subfield of AI where machines learn from data to perform specific tasks
Deep Learning	A subset of ML that uses artificial neural networks with many layers to extract complex patterns from data
Generative AI	An application of ML that focuses on creating new content
Foundation Models	Powerful ML models trained on massive amounts of unlabeled data, allowing them to develop a broad understanding of the world
Large Language Models (LLMs)	A type of foundation model that is designed to understand and generate human language

1.2 The AI Hierarchy

Level	Definition	Examples
AI	Broadest field — machines that simulate human intelligence	Chess engines, rule-based expert systems
Machine Learning	Subset of AI — learns patterns from DATA without explicit programming	Spam filters, recommendation engines
Deep Learning	Subset of ML — uses neural networks with many layers	Image recognition, speech-to-text
Generative AI	Subset of DL — CREATES new content: text, images, code, audio, video	Gemini, ChatGPT, DALL-E, Stable Diffusion

EXAM TIP ■ AI > ML > Deep Learning > Generative AI. Each is a SUBSET of the one above. Generative AI CREATES. Discriminative AI CLASSIFIES.

1.2 The Gen AI Landscape — 5 Layers

EXAM GUIDE ■ You must know these 5 layers and how Google products map to each one.

Layer	What It Is	Google Examples
1. Infrastructure	Compute hardware that powers AI workloads	TPUs, GPUs, Google Cloud compute

Layer	What It Is	Google Examples
2. Models	Pre-trained foundation models	Gemini, Gemma, Imagen, Veo
3. Platforms	Tools to build, tune, deploy AI	Vertex AI, Google AI Studio
4. Agents	AI that can take actions and use tools	Vertex AI Agent Builder, ADK, Agentspace
5. Applications	End-user facing AI solutions	Gemini for Workspace, NotebookLM, Gemini App

1.3 Google's Foundation Models

EXAM GUIDE SPECIFIC ■ The exam explicitly tests Gemini, Gemma, Imagen, and Veo. Know the modality and use case of each.

Model	Modality	Primary Use Case	Key Fact
Gemini	Multimodal (text+image+audio+video+code)	Google's flagship model for most tasks	1.5 Pro = 1M token context window
Gemma	Text (open-source)	Lightweight, open models for LOCAL deployments and specialized AI applications	Open weights — run locally, on-prem, or on-device without sending data to cloud
Imagen	Image generation	High-quality image creation from text prompts	Google's diffusion-based image model
Veo	Video generation	AI video creation from text or image prompts	State-of-the-art video generation model

1.4 Types of Generative AI Models

Model Type	What It Does	When to Use
Large Language Model (LLM)	Generates, summarises, translates text; writes code	Text tasks — Gemini Pro, Claude, GPT-4
Diffusion Model	State-of-the-art image/video generation (reverses noise process)	Image generation — Imagen, Stable Diffusion
Multimodal Model	Processes text + images + audio + video in ONE model	Cross-modal tasks — Gemini 1.5 Pro
Encoder-only Transformer	Text classification, sentiment analysis, NER	BERT — understanding tasks, NOT generation
Decoder-only Transformer	Text/code generation — most modern LLMs	Gemini, GPT — all generation models

1.5 Foundation Model Limitations

KEY FACT ■ Data Dependency | Knowledge Cutoff | Bias | Fairness | Hallucinations | Edge Cases. These are the exact 6 listed in the official Google certification course.

Limitation	What It Means	How to Mitigate
1. Data Dependency	Model quality is ENTIRELY dependent on the quality, diversity, and volume of training data. Garbage in = garbage out	Use high-quality, diverse, well-curated training datasets. Validate data before training
2. Knowledge Cutoff	No awareness of events AFTER training data end date. Model cannot answer questions about recent events	RAG with live/real-time data sources. Google Search grounding
3. Bias	Patterns of unfair preference or prejudice in training data are learned and can be amplified in outputs	Diverse training data, bias audits, human review, bias testing across demographics
4. Fairness	Model may produce systematically different quality or accuracy of outputs for different demographic groups	Fairness testing across groups, diverse evaluation datasets, responsible AI review gates
5. Hallucinations	Model generates confident but FACTUALLY WRONG or completely fabricated information	RAG/grounding to anchor responses in verified real data. HITL review for high-stakes outputs
6. Edge Cases	Model performs poorly on rare, unusual, or out-of-distribution inputs not well represented in training data	Extensive testing, adversarial testing, human-in-the-loop for unusual inputs

MITIGATION TOOLKIT ■ Official mitigations: Grounding | RAG | Prompt Engineering | Fine-Tuning | Human-in-the-Loop (HITL). Know which technique addresses which limitation.

1.6 How to Choose a Foundation Model

EXAM TIP ■ Know these 8 factors for evaluating and selecting a foundation model for a business use case.

Factor	What to Evaluate	Business Example
Modality	What data types the model handles: text, image, audio, video, code	Legal contract analysis needs text model; visual search needs image model
Context Window	How much information the model can process in one request	Analyzing 100-page contracts needs large context window (Gemini 1.5 Pro)
Security	Data residency, access controls, compliance certifications	Healthcare or finance needs HIPAA/GDPR-compliant model hosting
Availability & Reliability	SLA uptime guarantees, regional availability, rate limits	Mission-critical apps need 99.9%+ SLA and multi-region support

Factor	What to Evaluate	Business Example
Cost	Per-token pricing, training costs, inference costs at scale	High-volume apps must model cost per query to ensure ROI
Performance	Response quality, speed/latency, accuracy benchmarks	Customer-facing chat needs fast latency; research tasks can tolerate delay
Fine-Tuning	Whether the model supports domain-specific fine-tuning	Legal AI may need fine-tuning on case law and legal terminology
Customization	Ability to adjust behavior, system prompts, safety settings	Enterprise needs custom personas, output constraints, topic restrictions
Ease of Integration	How easily the model connects with existing systems, SDKs, APIs, and workflows	Low-friction integration reduces time-to-value and engineering costs

1.7 Machine Learning Approaches

Approach	Training Data	Primary Use Cases
Supervised Learning	Labeled data — input + correct output pairs	Classification, regression — fraud detection, price prediction
Unsupervised Learning	Unlabeled — model discovers patterns itself	Clustering, anomaly detection, dimensionality reduction
Reinforcement Learning	Agent learns via rewards/penalties (trial & error)	Game AI, robotics, RLHF for aligning LLMs with human values
Self-supervised	Labels derived from the data itself	How LLMs are pre-trained — predict next token from context

1.8 The ML Lifecycle — Official Stage Order

EXAM TIP ■ Data Ingestion -> Data Preparation -> Model Training -> Model Deployment -> Model Management. Know the Google Cloud tool for each stage.

1	Data Ingestion	Collect data from databases, files, APIs. Goal: relevant, quality data for training
2	Data Preparation	Clean, transform, normalize, remove duplicates. Split into train (70-80%) / test (20-30%)
3	Model Training	Feed training data to algorithm — model learns patterns. Validate to fine-tune parameters
4	Model Evaluation	Use test data to assess performance. Metrics: accuracy, precision, recall, F1-score
5	Model Deployment	Deploy trained model to production — serves real-time or batch predictions via API
6	Model Management	Continuous monitoring for drift, performance degradation, bias. Retrain as needed

1.9 Data Concepts & Quality

Concept	Definition
Structured Data	Predefined schema — tables, rows, columns. SQL databases, spreadsheets. Easy to query
Unstructured Data	No predefined schema — emails, images, PDFs, videos, social posts. ~80% of enterprise data
Labeled Data	Has correct answer tags. Required for supervised learning. Expensive and time-consuming to create
Unlabeled Data	No tags. Used in unsupervised/self-supervised learning. LLM pre-training uses this
Data Quality (7 dimensions)	Accuracy, Completeness, Consistency, Relevance, Availability, Cost, Format — all 7 impact model performance
Data Accessibility	Data for model training needs to be readily available, usable, and in the proper format
Business Use Case Verbs	Gen AI can CREATE (new content), SUMMARIZE (condense), DISCOVER (find patterns), AUTOMATE (replace manual steps)

DOMAIN 2 | Google Cloud's Generative AI Offerings (~35% of exam — LARGEST)

BIGGEST DOMAIN ■ 35% of the exam. Know Google product names, what each does, and WHEN to use each. Most marks won or lost here.

2.1 Vertex AI — Google's Unified AI Platform

- Google Cloud's **unified platform for ALL AI/ML work** — training, tuning, deploying, monitoring, agents
- Think of Vertex AI as the **umbrella** that contains all Google Cloud AI capabilities

Vertex AI Model Training Capabilities

Capability	Detail
Managed training environment	Fully managed infrastructure — no need to provision or manage compute yourself
Prebuilt containers	Ready-to-use containers for popular frameworks: TensorFlow, PyTorch, scikit-learn, XGBoost
Custom training jobs	Bring your own training code and run at scale with any framework
Model evaluation tools	Built-in tools to assess model quality, compare versions, and measure fairness
Powerful compute	TPUs and GPUs available to dramatically accelerate training workloads
AutoML	Minimal-code guided training — no need to write a training loop
Deployment scaling	Deploy models and automatically scale resources up or down based on demand

Vertex AI Component	What It Does
Vertex AI Studio	Browser-based playground to test prompts with Gemini in the Google Cloud console — production prompt engineering
Model Garden	Curated catalog: Google models (Gemini) + open-source (Llama, Mistral) + third-party. Test & deploy from one place
Agent Builder / ADK	Build conversational AI agents and multi-agent pipelines grounded in your company's data
Vertex AI Search	Enterprise search + recommendations over internal docs, websites, structured data (uses RAG internally)
RAG APIs	Prebuilt RAG with Vertex AI Search + standalone RAG APIs for custom retrieval pipelines
AutoML	Train custom ML models with minimal ML expertise — no manual architecture design

Vertex AI Component	What It Does
Vertex AI Pipelines	Orchestrate end-to-end ML workflows: data prep -> train -> evaluate -> deploy
Model Registry	Version control for trained models — track, compare, deploy specific versions
Feature Store	Centralised repository for ML features — consistent across training and serving

2.2 Gemini Model Family

Model	Designed For	Key Characteristics
Gemini Ultra / 1.0 Ultra	Most complex, highest-capability tasks	Largest model, top quality, highest cost. Advanced reasoning
Gemini 1.5 Pro	Long context + multimodal production tasks	1 MILLION token context window. Handles text+images+audio+video+code
Gemini 1.5 Flash	Balanced performance & cost	Fast, cost-effective general-purpose model for most business tasks
Gemini 1.0 Pro	General business use cases	Standard quality-cost balance for most enterprise needs
Gemini Nano	ON-DEVICE — mobile / edge devices	Runs on Pixel phones offline. No internet required. Privacy-preserving

EXAM TIP ■ Gemini Nano = on-device (Pixel phones, no internet). Gemini 1.5 Pro = 1M token context, natively multimodal. Long docs or multimodal = Gemini 1.5 Pro.

2.3 Google's Prebuilt AI Products

Product	Primary User	Key Use Case & Features
Gemini App (gemini.google.com)	All users (consumer + enterprise)	Google's AI assistant. Gemini Advanced = most capable tier. Gems = custom AI personas for specific tasks
Gemini Enterprise	Enterprise teams	Customized search and conversation agents that access and understand data from internal sources — integrated into internal websites or dashboards
Google AI Studio (aistudio.google.com)	Developers (free tier)	Free browser-based Gemini prototyping. Get API keys, test prompts, no cost to start
Vertex AI Studio	Enterprise developers	Production prompt engineering within Google Cloud security perimeter
Gemini Code Assist	Software developers	AI coding assistant in IDEs (VS Code, IntelliJ). Code gen, review, debug, explain
Gemini for Workspace	All business users	AI in Gmail, Docs, Sheets, Slides, Meet — draft, summarise, create, analyse

Product	Primary User	Key Use Case & Features
Google Agentspace	Enterprise employees	AI assistant hub: NotebookLM API + multimodal search + custom agent capabilities connecting to company apps
NotebookLM	Knowledge workers	AI-powered note-taking and Q&A; over your own uploaded documents. Source-grounded responses
Cloud Natural Language API	Developers (quick start)	Pre-built NLP: sentiment, entity extraction, classification — NO training needed
Cloud Vision API	Developers (quick start)	Pre-built image analysis: object detection, OCR, face detection — NO training needed
Speech-to-Text / Text-to-Speech	Developers	Voice transcription and synthesis — used in customer service and contact centre solutions
Document AI API	Developers	Intelligent document processing — extract structured data from PDFs, forms, contracts
Translation API	Developers	Text translation across 100+ languages — powers multilingual customer experiences

2.4 Google AI Studio vs Vertex AI Studio

EXAM TIP ■ Know WHEN to use Vertex AI Studio versus Google AI Studio. This is a likely exam question.

2.5 What is a Gen AI Agent?

OFFICIAL DEFINITION ■ An application that OBSERVES the world and ACTS on it using its tools to achieve goals.

What Agents Can Do:

Capability	Description	Example Use Case
Understand natural language	Interpret user intent from conversational input, not just keywords	Customer asks: find all overdue invoices from Q3
Automate complex tasks	Break multi-step tasks into actions, call tools, execute in sequence	Research -> draft -> review -> send a report automatically
Personalisation	Adapt responses and actions based on user context, history, preferences	Recommend products based on past purchases and browsing behaviour

Agent Application Areas:

- **Customer service:** Handle inquiries, resolve issues, escalate to humans when needed
- **Employee productivity:** Research, summarise, draft, schedule, and automate internal workflows
- **Creative tasks:** Generate content, brainstorm ideas, produce images or video on demand

Two Types of Agent Workflows

Conversational Agent Workflow	Workflow Agent Workflow
Trigger: User types or speaks a message	Trigger: Form submission, file upload, or system event
Understand: AI interprets meaning and intent	Understand: AI defines the steps needed to complete the task
Call Tool: Searches web, queries databases, triggers actions	Call Tool: Executes integrations, data transformations, API calls
Generate Response: Produces a relevant conversational answer	Generate Result: Compiles the final output or report
Deliver: Responds to user in real-time conversation	Deliver: Sends via email, dashboard update, or database write

Google AI Studio	Vertex AI Studio
Free, public browser tool at aistudio.google.com	Inside Google Cloud Console — requires GCP project
Developers prototyping and experimenting	Enterprise teams building production AI applications
Personal API keys, no enterprise controls	Full enterprise security: VPC-SC, IAM, audit logging, CMEK
No billing integration — limited to free tier quotas	Integrated with GCP billing, SLAs, and compliance frameworks
Use for: learning, personal projects, hackathons, prototypes	Use for: production apps, regulated industries, enterprise deployments

2.5 Customer Engagement Suite

EXAM TIP ■ Know the Customer Engagement Suite. Know what each component does and how they work together.

Component	What It Does
Customer Experience Agent Studio	CX agents act as effective chatbots to your customers — built on Contact Center as a Service (CCaaS)
Agent Assist	Real-time AI suggestions to human agents during live customer calls — reduces handle time
Customer Experience Insights	Gain insights into all your communications with customers — trends, compliance gaps, coaching
CCaaS (Contact Center as a Service)	Enterprise-grade contact center solution native to the cloud — the platform CX Agent Studio is built on

Agentspace vs NotebookLM — Know the Difference

Feature	NotebookLM	Agentspace
Purpose	Deep dive into specific documents you upload	Enterprise AI assistant across ALL connected business systems
Data Scope	Only user-provided source documents	All connected business systems — Drive, Jira, Salesforce, etc.
Integration	Standalone — can connect to NotebookLM Enterprise	Unified search and automation across the entire organisation
Best For	Researchers, students, analysts studying a document set	Employees needing AI assistance across all their work tools

2.6 How AI Agents Use Tools

EXAM TIP ■ Agents interact with the external environment through 4 types of tools. Know what each type does.

Tool Type	What It Does	Example
Extensions	Allow agents to call external APIs and services to take real-world actions	Book a calendar event, send an email, call a REST API
Functions	Custom code the agent can execute to process data or run logic	Calculate pricing, transform data, run a custom algorithm
Data Stores	Connect agent to knowledge bases for RAG-based information retrieval	Company policy docs, product catalogue, FAQ database
Plugins	Pre-built integrations with third-party services	Salesforce, Jira, Workday, ServiceNow integrations

2.7 Types of AI Agents

OFFICIAL COURSE ■ Course 5 distinguishes two agent types. The exam tests your ability to identify which type fits a given scenario.

Agent Type	How It Works	Best For
Deterministic (Traditional)	Built with predefined paths and actions. Follows fixed decision trees and scripted flows	Structured tasks with known, finite outcomes — e.g., order status lookup, password reset
Generative	Defined with natural language using LLMs to give a real conversational feel	Open-ended tasks requiring reasoning and flexibility — e.g., complex customer support, research

Agent Architecture — 4 Components



REASONING LOOP ■ The reasoning loop is the agent's brain — it uses ReAct, CoT, or Metaprompting to decide what action to take next, observe the result, and continue until the goal is met.

2.8 Planning Your Gen AI Strategy

OFFICIAL COURSE ■ Course 5 closes with this 6-step Gen AI strategy framework. Know all 6 steps in order.

1	Establish a clear vision	Define what you want AI to achieve for your organisation — tie to business goals
2	Prioritise use cases	Identify highest-value AI use cases — score by business impact vs implementation effort
3	Invest in capabilities	Build the skills, data infrastructure, and tooling needed to execute on AI use cases
4	Manage change	Communicate, train staff, address concerns — technology alone doesn't drive adoption
5	Measure value	Track business outcomes (time saved, errors reduced, revenue impact) against baseline KPIs
6	Champion responsible AI	Embed Secure AI and Ethical AI practices throughout all AI initiatives from day one

2.10 Vertex AI MLOps Tools

EXAM TIP ■ Know the purpose of each Vertex AI MLOps tool. These are frequently tested in the context of model management questions.

MLOps Tool	Purpose
Feature Store	Share and serve ML features consistently across training and serving — prevents feature skew
Model Registry	Track changes, manage versions, and deploy specific model versions to production
Model Evaluation	Compare model performance across versions — accuracy, precision, recall, fairness metrics
Vertex AI Pipelines	Automate and orchestrate end-to-end ML workflows from data prep through deployment
Model Monitoring	Detect performance degradation, data drift, and prediction drift in production models

2.9 Google's Infrastructure Advantage

- **TPUs (Tensor Processing Units):** Custom Google ASICs purpose-built for matrix ops in ML. Powers Gemini training AND Vertex AI customer workloads
- **GPUs:** NVIDIA A100/H100 available on Vertex AI for training and inference
- **Full-stack advantage:** Research (DeepMind) + Models (Gemini) + Infrastructure (TPUs) + Cloud Platform (Vertex AI)
- **Enterprise-ready:** Security (VPC-SC, CMEK) + Compliance (SOC2, HIPAA, GDPR) + SLAs + Enterprise support

CRITICAL FACT ■ Customer data sent to Vertex AI APIs is NEVER used to train Google's foundation models. This is Google's enterprise data privacy guarantee — critical for regulated industries.

DOMAIN 3 | Techniques to Improve Gen AI Output (~20% of exam)

3.1 RAG — Retrieval-Augmented Generation

KEY TECHNIQUE ■ RAG is the most heavily tested technique in Domain 3. Connects LLM to private, live data. Solves hallucination + knowledge cutoff simultaneously.

How RAG Works — 4 Steps:

1	Retrieval	The LLM retrieves relevant information from external sources using tooling (vector search on embedded documents)
2	Augmentation	The retrieved information is incorporated into the prompt to the LLM as additional context
3	Generation	The LLM processes the augmented prompt and generates a response grounded in retrieved content
4	Iteration (optional)	The LLM can iterate on the retrieval process as necessary — re-retrieve if the first pass is insufficient

3.2 Grounding — Three Types

EXAM TIP ■ The exam distinguishes 3 grounding types. Know which to use for each scenario.

Benefits of Grounding

Benefit	Detail
Reduces Hallucinations	Anchors model responses in real, verified data instead of relying on training memory
Cites Sources	Responses include citations so users can verify information — builds trust
Confidence Scores	Grounding returns confidence scores so users know how strongly the response is supported

Grounding Type	Data Source	Google Product	When to Use
First-party enterprise data	Your own company's documents, databases, knowledge bases	Vertex AI Search data stores / RAG APIs	Internal policy Q&A, product support, compliance chatbots
Third-party data	Partner or licensed external data sources	Vertex AI Search with third-party connectors	Enriching responses with partner or industry data
World data (Google Search)	Real-time public web information	Grounding with Google Search in Vertex AI	Questions about current events, public knowledge, recent news

3.3 Fine-Tuning

- **PURPOSE:** **Changes the model's behavior, style, tone, or domain vocabulary** — NOT just knowledge access
- **How:** Re-trains model weights on a curated dataset of domain-specific input-output example pairs
- **Cost:** More expensive than prompting — but much cheaper than training from scratch

Use Fine-Tuning When...	Use RAG Instead When...
Need a specific output FORMAT the model keeps forgetting	Need access to current or live information
Teaching domain-specific jargon (medical, legal, finance)	Knowledge base is large or updates frequently
Consistent brand voice / communication style	Need citations and traceable sources
Model needs to adopt a specific behavior pattern	Want to avoid expensive retraining cycles

KEY INSIGHT ■ Fine-tuning changes **HOW** the model behaves. RAG changes **WHAT** information it can access. They are **COMPLEMENTARY** and often used together in enterprise deployments.

Fine-Tuning Use Cases

Use Case	Example
Content in a specific style	Generating marketing copy, reports, or summaries that match your brand voice exactly
Code generation in specific languages	Training on your internal codebase so the model writes code in your team conventions
Domain-specific translation	Fine-tuning on medical, legal, or financial terminology for accurate translations in niche fields

3.4 Human-in-the-Loop (HITL)

EXAM TIP ■ HITL is explicitly listed in the official exam guide as a mitigation technique alongside RAG, grounding, and prompt engineering.

- **What it is:** Keeping humans involved in the AI decision loop to review, validate, or correct model outputs before they take effect
- **Why it matters:** Even the best models benefit from human oversight — technology reduces error rate, humans handle residual risk

HITL Use Case	Why Human Oversight Is Needed
Content Moderation	Ensures accurate, appropriate filtering of user-generated content — automated filters alone miss nuanced cases

HITL Use Case	Why Human Oversight Is Needed
Sensitive Applications	Provides oversight in healthcare, finance, legal — errors have serious real-world consequences
High-Risk Decisions	Adds accountability for decisions with serious consequences (hiring, loans, medical treatment)
Pre-Generation Review	Validates AI outputs BEFORE they are deployed or sent to users — catches errors early
Post-Generation Review	Continuous human feedback after deployment to improve models over time

3.5 Prompt Engineering Techniques

Technique	What It Is	Best For
Zero-shot	No examples — just instruction. Relies entirely on model's pre-training	Simple, clear tasks the model already handles well
One-shot	ONE example of input->output in the prompt	Teaching basic format when zero-shot isn't precise enough
Few-shot	2-5 examples of input->output pairs in the prompt	Teaching specific OUTPUT FORMAT or domain-specific style consistently
Role prompting	Assign a persona: 'You are a senior legal analyst...'	Setting tone, expertise level, communication style for entire conversation
Prompt chaining	Output of one prompt becomes input of next prompt	Complex multi-step tasks: research -> draft -> review -> format
Chain-of-thought (CoT)	Ask model to reason step-by-step before final answer	Complex reasoning, maths, multi-step logic where errors compound
ReAct prompting	Reason + Act: model reasons, takes action (tool call), observes result, continues	Agent workflows where model must decide to use tools mid-task
System prompts	Instructions in system role — persona, rules, format, constraints	Consistent behavior and format across all user turns in production

3.6 Sampling Parameters — Control Model Behaviour

EXAM TIP ■ Know these sampling parameters: temperature, top-p, safety settings, output length. Know what each controls.

Parameter	Effect	Setting Guidance
Temperature	Randomness of output. 0=deterministic. 1.0=creative/random	0-0.2 for factual Q&A/code; 0.7-1.0 for creative writing

Parameter	Effect	Setting Guidance
Top-P (nucleus sampling)	Model considers tokens in top P probability mass	0.9 = diverse; 0.1 = very focused. Use with temperature
Top-K	Model considers only top K most likely next tokens	Lower K = more focused; Higher K = more variety
Max output tokens	Maximum response length (hard cap)	Set to prevent runaway responses or ensure concise outputs
Safety settings	Thresholds for filtering harmful content categories	Adjust per use case — stricter for public apps, relaxed for internal research tools
Stop sequences	Strings that immediately halt token generation	Useful for structured output (stop at , stop at 'END')

3.7 Embeddings & Vector Search

- **Embeddings:** Dense numerical vectors capturing SEMANTIC MEANING of text — similar meaning = close vectors in space
- **Enables semantic search:** 'debugging code' finds docs about 'fixing bugs' — meaning matters, not exact keywords
- **Vector databases** (Vertex AI Vector Search / Matching Engine): Efficient similarity search at massive scale
- **Foundation of all RAG systems:** Query is embedded, then compared against embedded document chunks to find relevant content

DOMAIN 4 | Business Strategies for Gen AI Solutions (~15% of exam)

4.1 Google's AI Principles — Memorise All 7

#	Principle	What It Means in Practice
1	Be socially beneficial	AI should benefit society — not just the company deploying it
2	Avoid creating / reinforcing unfair bias	Test AI for bias across all demographics, especially in high-stakes decisions
3	Be built and tested for safety	Safety testing before deployment — do no harm, anticipate potential misuse
4	Be accountable to people	Humans can understand, challenge, audit, and override AI decisions
5	Incorporate privacy design principles	Data minimisation, purpose limitation, user consent, GDPR/CCPA compliance
6	Uphold high scientific excellence	Peer-reviewed methods, transparent about model limitations and uncertainty
7	Available only for uses aligned with principles	Google will not build AI for weapons, surveillance abuse, or illegal activities

4.2 Responsible AI

Pillar	Definition	Key Details
Secure AI	Protecting your AI applications from harm	Covers adversarial attacks, prompt injection, data poisoning, model theft. SAIF is Google's framework for this
Ethical AI	Ensuring your AI applications don't cause harm and are used in an ethical manner	Covers fairness, bias, transparency, accountability, explainability, privacy across all AI use cases

KEY FACT ■ The official course defines Responsible AI as TWO pillars: Secure AI (protecting from harm) and Ethical AI (avoiding causing harm). Know this split.

Ethical AI — Key Dimensions

Dimension	Definition	Exam Relevance
Fairness	AI outputs must not produce systematically different quality/accuracy for different demographic groups	Critical in hiring, lending, healthcare, criminal justice decisions
Explainability	Users, auditors, and regulators can understand WHY the AI made a decision	Required in regulated industries — black box AI is often non-compliant

Dimension	Definition	Exam Relevance
Accountability	Humans can audit, challenge, and override AI decisions — paired with explainability	Official guide pairs accountability and explainability together
Transparency	Be clear when users interact with AI — never impersonate a human agent	User trust + legal compliance in many jurisdictions
Privacy	Minimise data collection. Protect user data. Obtain consent. Apply GDPR/CCPA.	Data anonymization and pseudonymization are key techniques

Privacy Techniques

Technique	What It Is	When to Use
Data Anonymization	Permanently removing all personally identifiable information (PII) from data	When data must be shared externally or used for training — re-identification impossible
Data Pseudonymization	Replacing PII with pseudonyms/tokens — original data kept separately and securely	When you need to re-link data for authorised purposes but protect it at rest

HIGH-STAKES AI ■ Hiring, medical diagnosis, legal decisions, loan approval: ALL require HUMAN OVERSIGHT + bias testing + fairness evaluation before deployment.

Vertex Explainable AI

- **What it is:** A Vertex AI tool that makes AI decision-making transparent and understandable
- **Debug errors:** Understand why a model produced a wrong output and identify root cause
- **Uncover hidden biases:** Surface biases that are not obvious from model accuracy metrics alone
- **Build user trust:** Provide explanations that auditors, regulators, and end users can understand

4.3 Legal Implications of AI

EXAM TIP ■ AI development is governed by evolving legal frameworks. Know the 4 key legal areas.

Legal Area	What It Covers	AI Relevance
Data Privacy	GDPR, CCPA, and similar regulations governing collection and use of personal data	Training data and inference outputs must comply — data minimisation, consent, right to erasure
Non-Discrimination	Laws prohibiting discriminatory outcomes in hiring, lending, housing, healthcare	AI models trained on biased data can violate these laws — bias testing is legally required

Legal Area	What It Covers	AI Relevance
Intellectual Property	Copyright, patents, and ownership of AI-generated content	Who owns AI output? Was training data used without licence?
Product Liability	Legal responsibility when AI systems cause harm or make consequential errors	Organisations deploying AI may be liable for damages caused by model errors

4.3 Google Secure AI Framework (SAIF)

EXAM TIP ■ Be able to explain the purpose and benefits of Google's Secure AI Framework (SAIF). This is a likely exam question.

- **What SAIF is:** Google's framework for building and deploying AI systems securely throughout their entire lifecycle
- **Purpose:** Addresses AI-specific security risks that traditional security frameworks don't cover — like model manipulation, training data attacks, and inference-time threats
- **Scope:** Security applies at every stage — data collection, model training, deployment, monitoring, and updates
- **Key principle:** Secure-by-design — security built in from the start, not added as an afterthought

SAIF Component	What It Covers
Secure infrastructure	Google Cloud's secure-by-design compute, networking, and storage that AI workloads run on
Identity and Access Management (IAM)	Controls WHO can access AI models, training data, APIs, and outputs
Security Command Center	Unified threat detection and security monitoring for AI workloads and data pipelines
Workload monitoring	Ongoing monitoring of model behaviour in production to detect anomalies or adversarial activity
AI-specific threats	Prompt injection, adversarial attacks, data poisoning, model theft, jailbreaking — SAIF addresses all

4.4 Data Governance in AI

- Ensures data used in AI is **accurate, secure, private, compliant, and ethically used** throughout the AI lifecycle
- **Data lineage:** Track which data trained which model — essential for audits and regulatory compliance
- **GDPR/CCPA:** User consent for data use, right to erasure, data portability rights
- **Vertex AI data promise:** Customer data processed via Vertex AI is NEVER used to train Google's models
- **CMEK (Customer-Managed Encryption Keys):** Customer controls encryption keys — not Google
- **VPC Service Controls:** Network perimeter preventing data exfiltration from AI workloads
- **Data quality dimensions:** Completeness, Consistency, Relevance, Availability, Cost, Format

4.5 Google's Recommended AI Implementation Framework

1	Define the business problem	What specific outcome must improve? Set measurable KPIs BEFORE building anything
2	Identify key factors	Business requirements + technical constraints determine which AI solution is appropriate
3	Choose the right AI solution	Pre-built API vs. fine-tuning vs. RAG vs. custom model? Simplest approach that achieves the goal
4	Build responsibly	Apply bias testing, responsible AI checklist, SAIF security review, privacy assessment
5	Deploy and measure	A/B test, monitor KPIs vs. baseline — business outcomes, not just technical metrics
6	Iterate and scale	Use measurement results to improve. Scale successful pilots across the organisation

4.6 Measuring ROI — Business Value Metrics

- **Employee time saved** — hours/week x labour cost (most direct productivity ROI measure)
- **Task completion speed** — e.g., contract review: 4 hours -> 20 minutes (5x improvement)
- **Error rate reduction** — manual data entry errors reduced by X%
- **Customer satisfaction (CSAT/NPS)** improvement from AI-assisted support
- **Revenue impact** — personalised recommendations increasing conversion rate
- **Cost avoidance** — AI-first contact resolution reducing support ticket volume

EXAM TIP ■ ROI metrics = BUSINESS OUTCOMES. 'GPU hours consumed' or 'model accuracy on test set' are NOT ROI. 'Hours saved per employee per week' and 'error rate reduced by 40%' ARE ROI.

4.7 Change Management for AI Adoption

- Communicate the 'why' — connect AI adoption to company strategy and measurable employee benefits
- Provide training — employees need skills to use AI tools effectively and responsibly
- Address job displacement fears honestly — AI augments most roles, doesn't always replace them
- Identify AI champions — early adopters who evangelise within teams and share wins
- Start with low-risk pilots — build trust and evidence before high-stakes deployments
- Create feedback loops — employees report issues, suggest improvements, shape the rollout

EXAM SCENARIO ■ AI rollout failing despite good technology? The answer is almost always inadequate change management — missed training, poor communication, unaddressed employee concerns.

QUICK REFERENCE CHEAT SHEET | Last-Minute Revision

Product -> Use Case Decision Guide

Scenario / Need	Google Product	Why
Build & deploy custom AI/ML models at scale	Vertex AI	Unified ML platform — full lifecycle
Prototype prompts for FREE — no GCP account	Google AI Studio	Free browser playground for Gemini
Production prompt engineering within GCP	Vertex AI Studio	Enterprise security, audit logging, IAM
AI-powered enterprise search over internal docs	Vertex AI Search	Purpose-built RAG enterprise search
Build multi-agent AI workflows	Vertex AI Agent Builder / ADK	Multi-agent orchestration with tools
Code gen, debugging, review in IDE	Gemini Code Assist	Developer AI assistant for IDEs & CI/CD
AI in Gmail, Docs, Sheets, Slides, Meet	Gemini for Workspace	Embedded AI for all business users
Enterprise AI assistant for all employees	Agentspace	NotebookLM API + multimodal search + agents
Custom AI personas for specific tasks	Gems (in Gemini Advanced)	Role-specific AI assistants
Quick NLP — sentiment, entity extraction	Cloud Natural Language API	Pre-built API — no training needed
Customer service chatbots & voicebots	Conversational Agents (Dialogflow CX)	Customer Engagement Suite
Assist human agents on live calls	Agent Assist	Real-time AI suggestions for human agents
On-device AI — mobile, no internet	Gemini Nano	Runs on Pixel phones, privacy-preserving
Long documents, video, multimodal	Gemini 1.5 Pro	1M token context, natively multimodal
Open-source model, edge/on-prem deployment	Gemma	Open weights model, developer-friendly
Image generation from text	Imagen	Google's diffusion-based image model
Video generation from text/image	Veo	Google's video generation model

RAG vs Fine-Tuning vs Prompting vs HITL — When to Use

Prompting Only	RAG	Fine-Tuning	HITL
No new data needed Model already handles task Fastest/cheapest option	Need live/current data Need citations/traceability Large, frequently updated KB	Consistent style/tone/format Domain vocabulary & jargon Change model behavior	High-stakes decisions Edge cases & fairness risk Regulated environments

6 Official Foundation Model Limitations + Mitigations

Limitation	Best Mitigation
1. Data Dependency	High-quality, diverse, validated training data
2. Knowledge Cutoff	RAG with live data Google Search grounding
3. Bias	Diverse data + bias audits + demographic testing
4. Fairness	Fairness testing across groups + responsible AI review
5. Hallucinations	RAG/grounding to anchor in real data + HITL for high-stakes outputs
6. Edge Cases	Adversarial testing + HITL for unusual inputs + continuous monitoring

Key Glossary

Term	Definition
Token	~3/4 of a word on average. LLMs process text as tokens, not words
Embedding	Numerical vector capturing semantic meaning. Similar meaning = close vectors
Grounding	Connecting model to verified data sources — 3 types: 1st party, 3rd party, world/Search
HITL	Human-in-the-Loop — humans review/validate AI outputs for high-stakes decisions
SAIF	Google Secure AI Framework — security throughout the AI lifecycle (new in exam)
Context window	Max tokens model can process at once. Gemini 1.5 Pro = 1M tokens
RLHF	Reinforcement Learning from Human Feedback — aligns LLMs with human preferences
Gemma	Google's open-source foundation model for edge deployment — open weights
Imagen	Google's text-to-image diffusion model
Veo	Google's text-to-video generation model
Gems	Custom AI personas in Gemini Advanced — role-specific AI assistants
Agentspace	Enterprise AI hub: NotebookLM API + multimodal search + custom agents

Term	Definition
ReAct prompting	Reason + Act: model reasons then acts (tool call) then observes result iteratively
Data anonymization	Permanently removing PII — cannot be re-linked to individuals
Data pseudonymization	Replacing PII with tokens — original data kept separately, can be re-linked by authorised parties
CMEK	Customer-Managed Encryption Keys — customer, not Google, controls encryption

16 Last-Minute Exam Reminders

1. Official limitations = 6: Data Dependency, Knowledge Cutoff, Bias, FAIRNESS, Hallucinations, EDGE CASES
2. RAG = connects model to LIVE data + prevents hallucination + enables citations (3 benefits)
3. Fine-tuning = changes model BEHAVIOR/STYLE/TONE — not what it knows
4. HITL = official mitigation technique alongside RAG, grounding, prompt engineering, fine-tuning
5. Gemini Nano = on-device / Gemini 1.5 Pro = 1M token context, multimodal / Gemma = open-source edge
6. Imagen = image generation / Veo = video generation (both official exam models)
7. Google AI Studio = FREE prototype tool / Vertex AI Studio = enterprise production tool (know the difference)
8. Agentspace = NotebookLM API + multimodal search + custom agent capabilities
9. Domain 2 (GCP Offerings) = 35% of exam — know product names and use cases cold
10. Responsible AI: Fairness + Explainability + Accountability + Explainability + Transparency + Privacy + Safety
11. Google AI Principles = 7, starting with 'Be socially beneficial'
12. SAIF = Google's Secure AI Framework — security throughout AI lifecycle (new exam topic)
13. IAM + Security Command Center + workload monitoring = Google Cloud security tools for AI
14. Data anonymization (permanent) vs pseudonymization (reversible) — official exam distinction
15. ROI metrics = business outcomes (time saved, errors reduced) NOT GPU hours or model accuracy
16. NO prerequisites — exam is open to ALL job roles, technical or non-technical